

Stanford University

AA222/CS361: Engineering Design Optimization

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QUIZ 2

Due date: April 30, 2026 (10:20am Pacific)

Quizzes will be taken on Gradescope. You may consult any material (e.g., books, calculators, computer programs, and online resources), but you may not consult other people inside or outside of the class or large language models like ChatGPT. The quiz is designed to be completed in 60 minutes, but we will grant you 90 minutes total to complete and submit your quiz (including uploading any images, handling any logistical issues, etc). Partial credit may be awarded based on work shown. You can upload either a file or give a response in the free response field (L^AT_EX is possible using two \$ signs). The timing on Gradescope is a hard cutoff. Ed will not allow any public posts during that time. **Out of fairness to all students, only material submitted during the allowed time will be graded.**

Question 1. Stochastic Methods. We are using the cross entropy method to find the minimizer of $f(x) = (x - 4)^2$. Our proposal is a Gaussian distribution, and we use two elite samples to update the proposal at each iteration. In the initial population of designs, we have the samples 1, 5, -1, and 3. What is the mean of the proposal distribution at the next iteration?

Solution: The objective function is $f(x) = (x - 4)^2$. Evaluate the function at each sample:

$$f(1) = 9, \quad f(5) = 1, \quad f(-1) = 25, \quad f(3) = 1.$$

The two elite samples (with the smallest values) are $x = 5$ and $x = 3$. The updated mean is

$$\mu = \frac{5 + 3}{2} = 4.$$

Question 2. Population Methods. We are using particle swarm optimization to find the minimizer of $f(\mathbf{x}) = x_1^2 + x_2^2$. For a particular particle in our population, its current position vector is $[4, 4]$, its current velocity vector is $[2, -2]$, and its individual best position found so far is $[2, 0]$. The global best position found by the entire swarm is $[0, 0]$. We are using an inertia weight of 0.5, and both momentum coefficients (c_1 and c_2) are set to 1. Assuming the random acceleration weights for this step happen to be sampled as $[0.5, 0.5]$ and $[0.5, 0.5]$ for both the individual and global acceleration terms, what are the particle's velocity and position vectors at the next iteration? For both vectors, provide two digits after the decimal.

Solution: The update equations (as implemented in the code) are:

$$\mathbf{x}_{t+1}^{(i)} \leftarrow \mathbf{x}_t^{(i)} + \mathbf{v}_t^{(i)},$$

$$\mathbf{v}_{t+1}^{(i)} \leftarrow w\mathbf{v}_t^{(i)} + c_1\mathbf{r}_1 \odot (\mathbf{x}_{\text{best}}^{(i)} - \mathbf{x}_{t+1}^{(i)}) + c_2\mathbf{r}_2 \odot (\mathbf{x}_{\text{best}} - \mathbf{x}_{t+1}^{(i)}).$$

Using $w = 0.5$, $c_1 = 1$, $c_2 = 1$, $\mathbf{r}_1 = [0.5, 0.5]$, and $\mathbf{r}_2 = [0.5, 0.5]$.

Position update:

$$\mathbf{x}_{t+1}^{(i)} = [4, 4] + [2, -2] = [\mathbf{6.00}, \mathbf{2.00}]$$

Velocity update (using updated position):

$$\begin{aligned} \mathbf{v}_{t+1}^{(i)} &= 0.5[2, -2] + 1 \cdot [0.5, 0.5] \odot ([2, 0] - [6, 2]) + 1 \cdot [0.5, 0.5] \odot ([0, 0] - [6, 2]) \\ &= [1, -1] + [-2, -1] + [-3, -1] = [\mathbf{-4.00}, \mathbf{-3.00}] \end{aligned}$$

Other Accepted Solutions:

$$\mathbf{x}_{t+1}^{(i)} = [6.00, 2.00], \quad \mathbf{v}_{t+1}^{(i)} = [-2.00, -5.00]$$

$$\mathbf{x}_{t+1}^{(i)} = [2.00, -1.00], \quad \mathbf{v}_{t+1}^{(i)} = [-2.00, -5.00]$$

Question 3. Constraints. Suppose we have a constrained optimization problem where the objective is $x^2 + x + 3$ subject to $0 \leq x \leq 1$. Convert this problem into an equivalent unconstrained optimization problem, and write down the resulting objective function.

Solution: We remove the constraint $0 \leq x \leq 1$ using the transformation

$$x = \frac{b+a}{2} + \frac{b-a}{2} \cdot \frac{2\hat{x}}{1+\hat{x}^2}.$$

With $a = 0$ and $b = 1$, this becomes

$$x = \frac{1}{2} + \frac{\hat{x}}{1+\hat{x}^2}.$$

Substituting into the objective, the equivalent unconstrained optimization problem is as follows where $\hat{x} \in \mathbb{R}$ without constraints:

$$\min_{\hat{x}} \left(\frac{1}{2} + \frac{\hat{x}}{1+\hat{x}^2} \right)^2 + \left(\frac{1}{2} + \frac{\hat{x}}{1+\hat{x}^2} \right) + 3.$$

Question 4. Duality. Suppose we have a constrained optimization problem where the objective is $x^2 + x + 3$ subject to $0 \leq x \leq 1$. Write down the simplified dual form of this optimization problem expressed only in terms of the dual variables.

Solution: The constraints can be written as $-x \leq 0$ and $x - 1 \leq 0$. The Lagrangian is

$$\mathcal{L}(x, \mu_1, \mu_2) = x^2 + x + 3 + \mu_1(-x) + \mu_2(x - 1), \quad \mu_1, \mu_2 \geq 0.$$

$$\mathcal{L}(x, \mu_1, \mu_2) = x^2 + (1 - \mu_1 + \mu_2)x + 3 - \mu_2.$$

The primal form can be written as

$$\min_x \max_{\mu_1, \mu_2 \geq 0} \mathcal{L}(x, \mu_1, \mu_2),$$

and the dual form can be written as

$$\max_{\mu_1, \mu_2 \geq 0} \min_x \mathcal{L}(x, \mu_1, \mu_2).$$

Minimizing over x :

$$\begin{aligned} \frac{d\mathcal{L}}{dx} &= 2x + (1 - \mu_1 + \mu_2) = 0 \\ \Rightarrow x^* &= -\frac{1 - \mu_1 + \mu_2}{2} \end{aligned}$$

Substituting back,

$$\begin{aligned} g(\mu_1, \mu_2) &= \mathcal{L}(x^*, \mu_1, \mu_2) \\ &= \left(-\frac{1 - \mu_1 + \mu_2}{2} \right)^2 + (1 - \mu_1 + \mu_2) \left(-\frac{1 - \mu_1 + \mu_2}{2} \right) + 3 - \mu_2 \\ &= \frac{(1 - \mu_1 + \mu_2)^2}{4} - \frac{(1 - \mu_1 + \mu_2)^2}{2} + 3 - \mu_2 \\ &= -\frac{(1 - \mu_1 + \mu_2)^2}{4} + 3 - \mu_2. \end{aligned}$$

Thus, the simplified dual form is

$$\max_{\mu_1, \mu_2 \geq 0} -\frac{(1 - \mu_1 + \mu_2)^2}{4} + 3 - \mu_2.$$

Question 5. Linear Programming. Convert the following linear program to equality form:

$$\begin{aligned} &\text{minimize} && 2x_1 - x_2 \\ &\text{subject to} && x_1 + 3x_2 \leq 7, \\ &&& 2x_1 - x_2 \geq 4, \\ &&& x_2 \geq 0. \end{aligned}$$

Solution: Ensuring all decision variables are positively constrained:

Since x_1 is unrestricted in sign, write

$$x_1 = x_1^+ - x_1^-, \quad x_1^+, x_1^- \geq 0.$$

The problem becomes

$$\begin{aligned} & \text{minimize} && 2(x_1^+ - x_1^-) - x_2 \\ & \text{subject to} && (x_1^+ - x_1^-) + 3x_2 \leq 7, \\ & && 2(x_1^+ - x_1^-) - x_2 \geq 4, \\ & && x_1^+, x_1^-, x_2 \geq 0. \end{aligned}$$

Converting to equality form:

Introduce slack variables $s_1, s_2 \geq 0$:

$$\begin{aligned} & \text{minimize} && 2(x_1^+ - x_1^-) - x_2 \\ & \text{subject to} && (x_1^+ - x_1^-) + 3x_2 + s_1 = 7, \\ & && 2(x_1^+ - x_1^-) - x_2 - s_2 = 4, \\ & && x_1^+, x_1^-, x_2, s_1, s_2 \geq 0. \end{aligned}$$

Let

$$\mathbf{x} = \begin{bmatrix} x_1^+ \\ x_1^- \\ x_2 \\ s_1 \\ s_2 \end{bmatrix}, \quad \mathbf{x} \geq 0.$$

Then the problem in equality form is

$$\begin{aligned} & \text{minimize} && \mathbf{c}^T \mathbf{x} \\ & \text{subject to} && \mathbf{A}\mathbf{x} = \mathbf{b}, \\ & && \mathbf{x} \geq 0. \end{aligned}$$

$$\mathbf{c} = \begin{bmatrix} 2 \\ -2 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & -1 & 3 & 1 & 0 \\ 2 & -2 & -1 & 0 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}.$$

Question 6. Quadratic Programming. We have a quadratic program with objective $x_1^2 + 2x_1x_2 + 2x_2^2 - 4x_1 - 6x_2 + 5$. Convert this objective function into an equivalent least-squares objective function with the form $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|$. Write down $\mathbf{A}$ and $\mathbf{b}$ with two digits after the decimal for each numerical value.

Hint 1:

$$\begin{bmatrix} \sqrt{2} & 0 \\ \sqrt{2} & \sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{2} \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix}$$

Hint 2: When $ad - bc \neq 0$,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (1)$$

Solution:

The objective function is

$$f(\mathbf{x}) = x_1^2 + 2x_1x_2 + 2x_2^2 - 4x_1 - 6x_2 + 5.$$

Write it in the form

$$\frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{q}^T \mathbf{x} + 5, \quad \mathbf{Q} = \begin{bmatrix} q_1 & q_2 \\ q_3 & q_4 \end{bmatrix}.$$

Matching coefficients:

$$\mathbf{Q} = \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} -4 \\ -6 \end{bmatrix}.$$

Factor $\mathbf{Q} = \mathbf{U}^T \mathbf{U}$:

$$\mathbf{U} = \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{2} \end{bmatrix}, \quad \mathbf{U}^T = \begin{bmatrix} \sqrt{2} & 0 \\ \sqrt{2} & \sqrt{2} \end{bmatrix}.$$

Set $\mathbf{A} = \mathbf{U}$ and compute

$$\mathbf{b} = -\mathbf{U}^{-T} \mathbf{q} = -(\mathbf{U}^T)^{-1} \mathbf{q}.$$

Using Hint 2, invert $\mathbf{U}^T$:

$$(\mathbf{U}^T)^{-1} = \frac{1}{(\sqrt{2})(\sqrt{2}) - 0} \begin{bmatrix} \sqrt{2} & 0 \\ -\sqrt{2} & \sqrt{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \sqrt{2} & 0 \\ -\sqrt{2} & \sqrt{2} \end{bmatrix}.$$

Compute:

$$(\mathbf{U}^T)^{-1} \mathbf{q} = \frac{1}{2} \begin{bmatrix} \sqrt{2} & 0 \\ -\sqrt{2} & \sqrt{2} \end{bmatrix} \begin{bmatrix} -4 \\ -6 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -4\sqrt{2} \\ 4\sqrt{2} - 6\sqrt{2} \end{bmatrix} = \begin{bmatrix} -2\sqrt{2} \\ -\sqrt{2} \end{bmatrix}.$$

Therefore,

$$\mathbf{b} = -\mathbf{U}^{-T} \mathbf{q} = \begin{bmatrix} 2\sqrt{2} \\ \sqrt{2} \end{bmatrix}.$$

Final result (exact form):

$$\mathbf{A} = \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{2} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2\sqrt{2} \\ \sqrt{2} \end{bmatrix}.$$

Final result (decimal form):

$$\mathbf{A} = \begin{bmatrix} 1.41 & 1.41 \\ 0.00 & 1.41 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2.83 \\ 1.41 \end{bmatrix}.$$

Any other solutions which are scaled versions of $\mathbf{A}$ and $\mathbf{b}$ are also accepted.